limit area of the not-to-exceed zone for the applications of the engine.
3.3 The applicant's designated not-to-exceed zone can be defined by any mathematical formula(e), lists of coordinates or other method of defining the boundary. The not-to-exceed zone does not need to extend to the boundary of the limit area of the not-to-exceed zone.
3.4 For D2 and constant -speed E2 cycle certified engines, the not-to-exceed zone shall be a line of power greater than 25% at the nominal speed.
4 Determination of not-to-exceed emission limit value for E2, E3 and D2 test cycles
4.1 The not-to-exceed emission limit value at each NO x checkpoint shall be determined in accordance with the requirements of this section.
Note: If there is an auxiliary control device that causes a NOx discontinuity within the

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SOURCE: MEPC.397(83).pdf
test cycles
4.1 The not-to-exceed emission limit value at each NO x checkpoint shall be determined in accordance with the requirements of this section.
Note: If there is an auxiliary control device that causes a NOx discontinuity within the not-to-exceed zone, follow the additional procedure in section 6 to insert proxy NO x emission points to address the area(s) of discontinuity.
4.2 Interpolated NOx value at power between mode points as determined using formula (2):
= + ( )
( )
( ) (2)
where:
= Interpolated NOx value at power
= Measured point emission value according to formula (1) at nearest measured mode point at power below checkpoint power
= Measured point emission value according to formula (1) at nearest measured mode point at power above checkpoint power
= Power at checkpoint
= Power at mode point below checkpoint
= Power at mode point above checkpoint

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SOURCE: MEPC.397(83).pdf
= Power at mode point above checkpoint
RESOLUTION MEPC.397(83) (adopted on 11 April 2025)
AMENDMENTS TO THE TECHNICAL CODE ON CONTROL OF EMISSION OF NITROGEN OXIDES FROM MARINE DIESEL ENGINES (NOx TECHNICAL CODE 2008)- 16
4.3 Determine the not-to-exceed emission limit value at power between the mode points, to the Tier, as applicable.
.1 For Tier II
The not-to-exceed emission limit value at power is given by formula (3)
= 1.2 (3)
where:

= Not-to-exceed emission limit value at power
= Interpolated NO x value at power
.2 For Tier III
The not-to-exceed emission limit value, , at power shall be
either set by 3.1.4 of this Code or as determined in accordance with
formula (4), whichever is the lower.
is the lower of or
with:
= + 0.25 (4)
= 1.5 (5)
where:
= NOx cycle limit for engine
= the maximum mode point value for the engine according
to 3.1.4 of this Code
4.4 If the checkpoint power is on the propeller law curve for an E3 certified
engine or the nominal speed line for a constant-speed E2 or a D2 certified engine:
= (6)

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SOURCE: MEPC.397(83).pdf
to 3.1.4 of this Code
4.4 If the checkpoint power is on the propeller law curve for an E3 certified
engine or the nominal speed line for a constant-speed E2 or a D2 certified engine:
= (6)
For this situation the determination of the not-to-exceed emission limit value,
, is complete for that checkpoint. Otherwise, continue with 4.5.
4.5 For E3 and variable -speed application of the E2 cycle certified engines,
where the checkpoint power is located at a speed not on the line between the
measured mode points, carry out the additional procedure in 4.5.1 to 4.5.4.
.1 Determine the NO x limit at either edge of the not-to-exceed zone,
, for the selected checkpoint power along a line of constant
power, in accordance with formula (7):
RESOLUTION MEPC.397(83) (adopted on 11 April 2025)
AMENDMENTS TO THE TECHNICAL CODE ON CONTROL OF EMISSION
OF NITROGEN OXIDES FROM MARINE DIESEL ENGINES (NOx TECHNICAL CODE 2008)- 17 -
= 1.5 (7)
with:
=

(8)
where:
= NOx limit at edge of not-to-exceed zone
= NOx cycle limit for engine
= NO x specific emission value for the engine from 5.12.6.1 of
this Code

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SOURCE: MEPC.397(83).pdf
= 1.5 (7)
with:
=

(8)
where:
= NOx limit at edge of not-to-exceed zone
= NOx cycle limit for engine
= NO x specific emission value for the engine from 5.12.6.1 of this Code
.2 Determine the not-to-exceed emission limit value at a checkpoint power which is on the constant power line between the mode point line and the edge of the not-to-exceed zone in accordance with formula (9):
= + ( ) ()() (9)
with:
For tier II, = from formula (3)
For tier III, is from formula (4)
where:
= NOx limit at required checkpoint
= NOx limit at edge of not-to-exceed zone
= Speed at required checkpoint
= Speed at edge of applicants selected not-to-exceed zone at checkpoint power (may be on lower or higher side of mode line as required for value of)
= Speed on measured mode line at selected power
For engines certified to the E2 test cycle, the speed on the measured mode line, , is the nominal speed.
For engines certified to the E3 test cycle, the speed on the measured mode line, , is determined by the cube law propeller curve:
= .

3
(10)

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SOURCE: MEPC.397(83).pdf
line, , is the nominal speed.
For engines certified to the E3 test cycle, the speed on the measured mode line, , is determined by the cube law propeller curve:
= .

3
(10)

where:
= Rated speed from 1.3.12 of this Code
= Power at checkpoint
= Rated power from 1.3.11 of this Code
.3 Determine the not-to-exceed emission limit value at power as applicable:

.1 For Tier II
The NOx limit is the interpolated result:
= (11)
.2 For Tier III
The not-to-exceed emission limit value at power shall be
set by 3.1.4 of this Code or as determined in accordance
with 4.5.3.1 whichever is lower:
is the lower of or
5 Determination of not-to-exceed emission limit value for the C1 test
cycle
5.1 For the C1 test cycle within the limit area of the not-to-exceed zone,
screening is conducted between the measured mode points of 100%, 75%, and 50%
load at both intermediate speed (mode points 5, 6 and 7 respectively) and rated speed

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SOURCE: MEPC.397(83).pdf
cycle
5.1 For the C1 test cycle within the limit area of the not-to-exceed zone,
screening is conducted between the measured mode points of 100%, 75%, and 50%
load at both intermediate speed (mode points 5, 6 and 7 respectively) and rated speed
(mode points 1, 2 and 3, respectively).
This creates two zones, Zone A and Zone B, where double linear interpolation or